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MINI-REVIEW

Antimicrobial resistance: new insights and therapeutic implications

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Abstract

Antimicrobial resistance has not been a new phenomenon. Still, the number of resistant organisms, the geographic areas affected by emerging drug resistance, and the magnitude of resistance in a single organism are enormous and mounting. Disease and disease-causing agents formerly thought to be contained by antibiotics are now returning in new forms resistant to existing therapies. Antimicrobial resistance is one of the most severe and complicated health issues globally, driven by interrelated dynamics in humans, animals, and environmental health sectors. Coupled with various epidemiological factors and a limited pipeline for new antimicrobials, all these misappropriations allow the transmission of drug-resistant organisms. The problem is likely to worsen soon. Antimicrobial resistance in general and antibiotic resistance in particular is a shared global problem. Actions taken by any single country can adversely or positively affect the other country. Targeted coordination and prevention strategies are critical in stopping the spread of antibiotic-resistant organisms and hence its overall management. This article has provided in-depth knowledge about various methods that can help mitigate the emergence and spread of antimicrobial resistance globally.

Key points

• *Overview of antimicrobial resistance as a global challenge and explain various reasons for its rapid progression.*
• *Brief about the intrinsic and acquired resistance to antimicrobials and development of antibiotic resistance in bacteria.*
• *Systematically organized information is provided on different strategies for tackling antimicrobial resistance for the welfare of human health.*

Keywords Antibiotic resistance · Horizontal gene transfer (HGT) · *M. tuberculosis* · Quorum-sensing · Acquired resistance · Biofilms · Intrinsic resistance · CRISPR-Cas system

Introduction

Antimicrobial-resistant (AMR) infections severely threaten human and animal health. Antimicrobials are drugs that are active against different pathogens, including bacteria (antibiotics), fungi (antifungals), viruses (antivirals), and parasites (including antimalarials) (Ghosh et al. 2019). Once the microorganisms causing infections (e.g., bacteria) possess the potential of survival even after their exposure to medicine that otherwise will typically kill or stop their growth, they are said to exhibit antimicrobial resistance (AMR) (Michael et al. 2021). In this context, antibiotic resistance in bacteria is a multifactorial process influenced by various factors, but the most being attributed to the overuse of antibiotics in humans and other animals (Jin et al. 2021). AMR has been observed way back since discovering the first antibiotic drug (Aslam et al. 2018). AMR is primarily caused by uncontrolled antibiotic usage, which is increasing at an alarming rate worldwide. But at the same time, we are deprived of new drugs to combat and challenge these new superbugs. This has presented a challenge to the scientific community and has fought an enemy with a largely depleted weapon (Cangini et al. 2020). The development of 'superbugs' including methicillin-resistant *Staphylococcus aureus* and extremely drug-resistant *Mycobacterium tuberculosis*, against which current antibiotics are least efficient and difficult to treat, is due to the emergence of resistance potential (Sheikh et al. 2022). The development of AMR in bacterial pathogens has resulted in an increased rate of mortality and morbidity across the globe (Akova 2016; Velez and Sloand 2016).

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Moreover, the early detection of causative microbial agents and their susceptibility patterns to antimicrobial drugs in bacteremia patients and patients with other infections is insufficient in various healthcare settings (Akova 2016; Sheikh et al. 2020). A substantial surge in resistance occurs due to poor disease control practices leading to the dissemination of resistant bacteria to different patients and the surrounding environment (Akova 2016). In the past, the main sites for spreading resistant infections were predominantly considered care settings and hospitals. Still, in the last decade, such resistant infections have been observed in the broader communities (Kraemer et al. 2019). In addition to the most commonly encountered antibiotic-resistant genes (ARGs) in clinical pathogens, all pathogenic, environmental, and commensal bacteria, bacteriophages, and mobile genetic elements from a repository of antibiotic-resistant genes from which pathogenic bacteria may gain resistance via horizontal gene transfer (HGT) (Von Wintersdorff et al. 2016). As observed in previous studies, the relevance of transduction and transformation with the emergence of AMR is significantly less. Still, essential discoveries recommend that their participation be more significant than previous studies (Nguyen et al. 2021). Hence, it is crucial to understand the level and the mechanism of antimicrobial resistance and its transmission to pathogenic bacteria to control the spread of these resistant genes. As a result of such a daunting scenario of resistance, we may lose the immense ground and breakthroughs we had achieved in the last century. Due to this, the entire world desperately needs antibiotics or antimicrobial stewardship programs intended to prevent antibiotic overuse and reduce antibiotic resistance (Akova 2016; Puvača and de Llanos 2021). In the battle against bacterial infections and other related diseases (e.g., fungal diseases), there are different obstacles, including the lack of effective drugs, the absence of effective counteracting steps, and the presence of only a few novel antibiotics in the clinical pipeline. Controlling such resistant pathogens requires developing new treatment methods and alternative antimicrobial therapies (Mühlen and Dersch 2015). Therefore, discovering various ways and strategies to reduce the growth and spread of antibiotic resistance poses a challenge to the research community and the management of public health at large (Chellat et al. 2016; Mir et al.). From this perspective, the global prevalence of antimicrobial resistance in general, and antibacterial resistance in particular, are discussed in the following sections of this article. In addition, a detailed overview of various novel strategies that could be used to mitigate AMR is also explored.

Emerging strategies to combat antibiotic resistance

Intrinsic resistance, alternatively referred to as innate or primary resistance, refers to the general insensitivity of a bacteria or any other microbe to a specific antimicrobial drug or class of drugs (Van Duijkeren et al. 2018). This is frequently a result of the absence or inaccessibility of target structures for specific antimicrobial agents, such as resistance to β-lactam antibiotics and glycopeptides in bacteria (*Mycoplasma* spp.) lacking a cell wall. Similarly, vancomycin resistance in Gram-negative bacteria is due to the inability of vancomycin to penetrate the outer membrane. Intrinsic resistance can also occur due to export systems or the secretion of species-specific inactivating enzymes by certain bacteria, for example, the AcrAB-TolC system or the production of AmpC β-lactamase by certain *Enterobacteriaceae*. Furthermore, some bacteria, such as *Enterococci*, can utilize exogenous folates and thus not rely on a functional folate synthesis pathway. Bacteria have an intrinsic resistance that is unique to each species or genus.

On the other hand, acquired resistance is a strain-specific feature caused by acquiring foreign resistance genes or mutational modifications of chromosomal target genes. This category may include mutations that upregulate the expression of multidrug transporter systems. There are three fundamental types of resistance mechanisms: (i) enzymatic inactivation of antimicrobials via disintegration or chemical modification, (ii) lowered intracellular accumulation of antimicrobials due to decreased influx and increased efflux of antimicrobials, (iii) overexpression of sensitive targets or the replacement of sensitive target structures with alternative resistant structures, as well as mutation, chemical modification, or protection of cellular target sites (Fig. 1).

Ongoing research projects and other continuous efforts are devoted to combating antibiotic resistance. Various methods have been developed to decrease resistance, but a specific class or single strain of resistant microbe has primarily been targeted. Here, several antibiotic resistance strategies are highlighted, instigating the development of systematic initiatives to combat antibiotic resistance. These new approaches show productivity and efficiency, with very little probability of causing drug resistance and paving the way for further breakthroughs in infection control drug development. Several methods have been adopted to decrease mutation supply: primarily drugs that inhibit or prevent mechanisms (e.g., Lex A, Mfd) responsible for the elevation in the rate of mutation. Secondly, drugs are responsible for inhibiting horizontal gene transfer through conjugative plasmids (Getino and De la cruz 2018; Ragheb et al. 2019).

Moreover, the production of such drugs is as laborious as that of new bonafide antibiotics, and thus their use is also undermined (Sommer et al. 2017). Besides, it is indistinct how in phase 2–3 trials, the clinical effectiveness of such types of drugs can be determined. Antimutagenesis drugs may be administered in conjunction with antibiotics in treated patients where the emergence

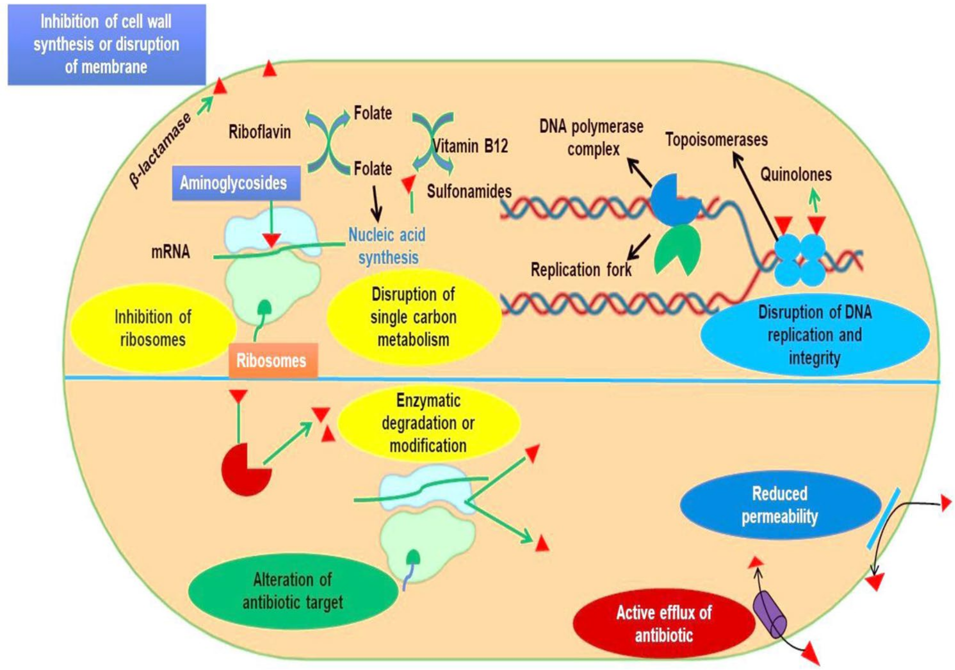

Fig. 1 Molecular mechanism of antibiotic resistance in bacteria. These mechanisms include enzymatic modification of the drug, modification of the antimicrobial target, and prevention of drug penetration or accumulation

of resistance is supposed to occur via mutation, as in the case of *M. tuberculosis* (Sheikh et al. 2020). For inhibitors of conjugation, since the development of resistant bacteria frequently takes place outside of treated patients via the horizontal gene transfer process, these antibiotics may be more suitable for research and treatment in conditions of high transfer risk, such as animal husbandry, as compared to human patients (Qadri et al. 2021; Tenover 2006) (Table 1).

Modulating human microbiome and antibiotic resistance

Human microbiomes are complex ecosystems that include bacteria, viruses, archaea, or eukaryotes co-evolving in an environment exposed to numerous selective pressures, including administration of antibiotics, diet, and lifestyle (Lagier et al. 2016; Peterson et al. 2009; Rolain 2013). According to multiple research works, antibiotic use has a variety of impacts on the gut microbiota. Still, it reduces the relative abundance of microorganisms in the gut and increases the number of resistance genes in most cases. Competition is fierce in this sympatric lifestyle, and one response devised by the organisms to survive is the synthesis of antibiotic molecules and antibiotic resistance genes (ARGs). This environment becomes a significant source of antimicrobial resistance genes (ARGs) for pathogenic bacteria, increasing the likelihood of infection from multidrug-resistant bacteria. Culture and metagenomics are two complementary ways of studying these microbiomes to understand the bacteria and ARGs found in the human body and the variables and factors that modulate their abundance and diversity. Most metagenomic functional selections are used to detect several hundred resistance genes in the microbiome of healthy people (Sommer et al. 2009). The microbiome is a source of antimicrobial resistance in host species.

Table 1 Major antibiotic families and their target of action in bacteria

Major antibiotic families and their target of action	
Target of action	Antibiotic families
Inhibition of cell wall synthesis	Penicillin's; cephalosporins; carbapenems; daptomycin; monobactams; glycopeptides
Inhibition of protein synthesis	Tetracyclines; aminoglycosides; oxazolidonones; streptogramins; ketolides; macrolides; lincosamides
Inhibition of DNA synthesis	Fluoroquinolones
Competitive inhibition of folic-acid synthesis	Sulfonamides; trimethoprim
Inhibition of RNA synthesis	Rifampicin
Other	Metronidazole

Administration of clindamycin, for example, decreases the relative abundance of anaerobic bacteria like *Bacteroides* while increasing the relative abundance of *Enterobacteriaceae* in infected adults (Pérez-Cobas et al. 2013).

Moreover, an essential role is played by animal and human opportunistic pathogens such as intestinal *Enterobacterales* or *Enterococcus*. Commensal bacteria that exhibit antibiotic resistance (e.g., β-lactamases-producing intestinal *Bacteroides*) mainly carry antibiotic degradation, thus reducing the selective power of drugs and protecting our endogenous microbiota from potential damage. In their specific biological niche, these species create a barrier against more widely spread resistance, which can be considered as a "resilience mechanism" that leads to "colonization-resistance" against drug-resistant pathogens (Ruppé et al. 2019). Therefore, commensal microbiota assumes a critical function in the battle against the dispersal of resistance. This attribute shall be considered before implementing strategies that will carry the targeted elimination of particular antibiotic-resistant bacteria. Faecal microbiota transplantation (FMT) is investigated and is well known to eliminate and destroy the organisms possessing multidrug resistance (Gopalsamy et al. 2018). Phage therapy can be valuable in wiping out drug-resistant and opportunistic pathogens, perpetuating dysbiosis (Kortright et al. 2019). Optimized use of pre-and/probiotics, in situ conjugation, and dietary interventions to modify the metabolic landscape of the gut (e.g., short-chain fatty acids) are the strategies for microbiome modulation (Goren et al. 2019; Ronda et al. 2019). Some approaches that carry microbiome manipulation are associated with risks that cannot be underestimated (DeFilipp et al. 2019).

Improving healthcare practices and enacting effective health policies

Overuse of antibiotics in an uncontrolled manner will finally weaken their efficacy against pathogens in humans. Conversely, more controlled and reasonable utilization of antibiotics will be significantly helpful in preserving the effectiveness of antibiotics in response to infectious diseases, ultimately resulting in the protection of health conditions of humans with significant benefits at the individual level and the community level (Baquero 2007). Usually, in several cases, the prescription of antibiotics for treating mild community-based infections, perhaps without sufficient awareness of such use, escalates worldwide antibiotic resistance. Non-antibiotic medication is significant for mild infections; for example, anti-inflammatory regimens, including ibuprofen, may be examined while utilizing biomarkers, such as lactate and pro-calcitonin, to screen patients for sepsis and bacterial infection (Baquero et al. 2015; Branche et al. 2019). Besides immunocompetent hosts with mild infections, bacteriostatic antibiotics are comparable to bactericidal ones, and short therapeutics courses are usually adequate (Nemeth et al. 2015). One more effective healthcare practice is immunization and vaccination of the population against viruses, including influenza, which decreases the host's susceptibility to infections spread by bacteria (Kash and Taubenberger 2015). The principle is based on three factors; the number of cases where diagnostic uncertainty favors the use of antibiotics is primarily reduced. Secondly, it reduces the number of co-infections secondary to antibiotic-curable infectious diseases (Mir et al. 2022a). Finally, the increased rate of transmission between the hosts of resistant bacterial pathogens followed by virus-caused infections is also avoided. Additionally, another approach is to minimize hospital stays; this will increase hospital turnover and the number of admitted patients, resulting in a decrease in the prevalence of antibiotic-resistant organisms at the local level (Lipsitch et al. 2000). The detection of bacteria-resistant to antibiotics can be carried out by the rapid diagnostic tools ahead of the admission of the new patient in the hospital in the high-risk areas of transmission (Mir et al. 2022b). This may result in containment steps for patients colonized by the antibiotic-resistant organism being quickly introduced.

Reduction of selective pressure and narrowing window of selection

To mitigate the development of antibiotic-resistant microbes, the most immediate and efficient solution is to decrease the relevant selective pressures, which gives them a fitness advantage. Therefore, recognizing certain factors whose reduction will be enormously successful in reducing the development of resistance will be very helpful and advantageous. Medical clinics and hospitals are the main centers of disseminating antibiotic resistance. There is the unnecessary use of antibiotics and the presence of hosts with a high degree of susceptibility to infections. However, to decrease the selective pressure for antibiotic resistance, judicious use of antibiotics in hospital settings and hospital hygiene are some of the essential and effective strategies (Baur et al. 2017). In humans, most antibiotics are used outside hospitals, as reported by some studies. Therefore, continuous monitoring of antibiotic use for most common conditions, including urinary tract and respiratory infections, is equally essential (Low et al. 2018). To this end, novel and rapid diagnostic tests that could differentiate bacterial respiratory and viral infections and effective drug susceptibility tests are needed. This would encourage the evidence-based choice of antibiotic drugs rather than empirical-based and more rapid de-escalation. In addition, the scientific community is currently challenged by the question to which degree the utilization of antibiotics in animals contributes to the emergence of resistance to human-related pathogens and, subsequently, the effect of decreased use of antibiotics in animals on human resistance (Dorado-García et al. 2016; Mughini-Gras et al. 2019). For biocides and metals, including silver, their part in driving antibiotic resistance is not determined; however, as per available laboratory experiments, these should also be used with care (Maillard 2018). Commonly, resistant microbes have an antibiotic concentration-dependent selective advantage only in a particular segment of antibiotic concentrations. As such, decreased concentration does not exhibit any effect, whereas very high concentrations may inhibit immune cells. This specified segment represents the selection window, and the longer these selective concentrations remain during infection or at the colonization site, the more the possibility of resistance development (Drlica and Zhao 2007; Hughes and Andersson 2012). When antibiotics with a prolonged half-life are taken, the probability of selection for resistant microbes increases. Pharmacokinetic/pharmacodynamics modeling and simulation must promote judicious antibiotic treatment to ensure safety and effectiveness in clinical practice. The characteristics of pharmacokinetic/pharmacodynamics associated with various drug and dosing regimens indicate that it is possible to narrow the selection window and reduce the development of antibiotic-resistant organisms.

A combinatorial approach of antibiotics, use of adjuvants, and antibacterial vaccination

The present situation demands better information and various advanced healthcare practices regarding combination therapy. Even though this type of combination therapy for most infections is not significantly effective compared to monotherapy with a single drug (Tyers and Wright 2019). However, the risk of resistance within an organism is reduced when combination therapy is used. When this type of combination therapy is used, variants that show resistance to one drug remain susceptible to another (vice versa) present in the combination. Tuberculosis infection treatment constitutes one of the cases of combination therapy, but it is also applicable to infections in immunosuppressed patients. Therefore, the combination of antibiotics should be considered one of the strategies for reducing the challenge of the emergence of antibiotic resistance and extending antibiotic life (Mebis et al. 1998; Tyers and Wright 2019). Given the profound interpretation of the range of drug resistance to individual drugs, the evolution-based use of antibiotic combinations should allow the identification of practical, successful, and less prone to resistance (Mir 2015). Besides, in different situations, single-antibiotics are inadequate and ineffective in killing specific susceptible pathogens because, under in vivo conditions, these pathogens are partly susceptible to them, typically due to slow growth and hetero-resistance (Mir 2022a, b).

In situations such as endovascular infections, endocarditis, etc., different antibiotics, including aminoglycosides, are usually added to target these types of dormant populations, thus leading to the generation of combination therapy that is more effective than single drugs alone. Moreover, antibiotic combinations must be chosen cautiously as per the particular clinical case (Masters et al. 2003). It is not assured that both (antibiotic molecules) will enter the same bacterial cell when the two antibiotic molecules are used in combination. Some in vivo studies observe that the de-combination of these drugs takes place up to a certain degree. Using nanocarriers as delivery vehicles for both drugs could compensate for and overcome this disadvantage (Walvekar et al. 2019). The recent advances in nanobiotechnology open a new window to combat antibiotic resistance. Nanoparticles combat antibiotic resistance by different mechanisms. Several nanoparticles have been developed that inhibit antibiotic resistance; these include nitric-oxide releasing nanoparticles (NO NPs), nanoparticles containing chitosan, and some metal-based nanoparticles. Nanoparticle-based drug delivery generally damages the DNA, RNA, and protein machinery of bacteria via the action of reactive nitrogen oxide intermediates (RNOS) (Qasim et al. 2014). Reactive nitrogen oxide intermediates also inhibit cellular respiration by inactivating zinc metalloproteins (Baptista et al. 2018). Furthermore, nanoparticle such as titanium oxide (TiO_2 NP) also stimulates an immune response (innate) in the human host (Baptista et al. 2018; Mir et al. 2021).

Antibiotic activity may be potentiated by various non-conventional antimicrobials, narrowing the selective window of resistance (Andersson et al. 2020). For instance, peptides (amphiphilic) or purified phage lysins make the bacterial cells more permeable, reduce the survival of resistant mutants (low-level), and reduce a cell population that may, with time, show acquired or mutational resistance. In addition, the antibiotic activity can be boosted by adjuvants which may enhance and promote the killing of the bacteria by facilitating the generation of the harmful reactive oxygen species (ROS) in the treated bacteria. Therefore, adjuvant therapy is an orthogonal approach to producing new antibiotics. Research is very operative in the domain of antibiotic hybrids, antibiotic enhancers, and synthetic constructs of the antibiotics that yield complementary antibiotic capability (Domalaon et al. 2018; Wright 2016).

Antibacterial vaccination

Vaccination against bacteria can be an exceptionally viable intervention against bacterial pathogens with antibiotic resistance (Jansen et al. 2018). This has been unambiguously shown in resistant *Streptococcus pneumonia*:

where conjugate vaccines (polysaccharide-protein), namely PCV7 and PCV13, have been successfully used. This has been achieved by targeting serotypes that showed resistance to macrolides, β-lactams, and fluoroquinolones. Another promising bio-conjugate vaccine for *E. coli* has been developed containing the O-antigens of four serotypes of *E. coli* (ExPEC4V). Furthermore, against *Salmonella*, oral vaccines are effective. In the future, similar types of vaccines can be produced to decrease gut colonization by resistant *Enterobacterales* (Begier et al. 2017; Huttner and Gambillara 2018; Wahid et al. 2016) (Table 2).

To combat the resistance, passive vaccination can also be used. The emergence of antibiotic resistance is more often found in immune-compromised patients and in situations where immune surveillance is missing (biofilms, catheters) (DeNegre et al. 2019). The above shows that immunity contributes significantly to checking and delaying antibiotic-resistant-organisms (Levin et al. 2017). In addition, applying a combination of antibodies (monoclonal and polyclonal) and antibiotics against a resistant clone of bacteria can also be beneficial. This may prove a successful strategy soon (Al-Hamad et al. 2011; Saylor et al. 2009). The recent approach to cutting-edge protein microarray technology may also lead to the production of new vaccines, as the humoral immune response to entire multidrug-resistant bacterial proteomes can be profiled by using this technique.

Screening of novel targets and development of antibiotics against them

An intriguing argument is to determine whether it is feasible to identify molecular markers that are less likely to lose effectiveness over time due to resistance and to pay more attention to such targets throughout the development of the drug. The most significant bacterial factors need to be identified for sustained drug efficacy. The ideal antimicrobial must possess a steep pharmacodynamic response in both drug-susceptible and drug-resistant bacteria. Thus, for the resistance to occur, there will be a high fitness cost for the ideal molecular marker (target) if it shows mutation and will also exhibit inefficient rate compensatory evolution (Qadri et al. 2021; Regoes et al. 2004). Lower-killing kinetics stress the residual population of bacteria that are usually still quite dense, and favor mutational variation or recruitment of resistance genes (Poole 2012). It has been found that no selective effect occurs by the concentration of antibiotics which are very low to show the effect on the bacterial population or by the concentration of antibiotics that has the potential to kill all the bacteria cells; thus, it is

Table 2 List of vaccine candidates in the pipeline (in US and EU) based on the centers for disease control and prevention (CDC) antibiotic-resistant threat list

Target	Clinical -stage pipeline				Vaccine in use (Licensed by FDA)
	Phase 1	Phase 2	Phase 3	Total	
Urgent threats					
Clostridium difficile	X	2	1	3	0
Carbapenem-resistant Klebsiella	X	X	X	0	0
Neisseria gonorrhoeae	X	X	X	0	0
Carbapenem-resistant *E. coli*	2	X	X	2	0
Serious threats					
Mycobacterium tuberculosis	1	4	X	5	1
Streptococcus pneumoniae	1	3	X	4	3
Candida	X	1	X	1	0
Salmonella typhi	X	X	X	X	2
Pseudomonas aeruginosa	X	1	X	1	0
Campylobacter	X	X	X	0	0
Shigella	X	X	X	0	0
Enterococcus	X	X	X	0	0
Acinetobacter	X	X	X	0	0
Non-typhoidal *Salmonella*	X	X	X	0	0
Concerning threats					
Staphylococcus aureus	3	1	X	4	0
Group A *Streptococcus*	X	X	X	X	0
Group B *Streptococcus*	X	1	X	1	0

In this table, “X” indicates that there are no candidates in the pipeline expected that the intermediate concentration of antibiotics is the selective window for the emergence of resistance. The selective window should be limited to the degree that the time spent within this selective window during treatment is very minute or negligible (Drlica and Zhao 2007; Sommer et al. 2017). New drugs that have been produced and do not cross-resist with the already existing antimicrobials should be equally active on microbes susceptible or immune to previous antibiotics (Gerding et al. 1991). The presumptions concerning resistance mediated by the horizontal gene transfer (HGT) are significantly different than those for resistance due to mutation. Several parameters for reducing the HGT include; (i) Whether genes that are responsible for conferring resistance to the selected molecules have evolved to exhibit resistance with some mutations or are naturally present in an organism, (ii) HGT rate, (iii) the fitness effects on the recipient bacteria after HGT, (iv) the ecological opportunity of HGT and (v) probability that the genes responsible for resistance mobilize between various HGT vectors with a varied tendency for horizontal gene transfer (Sommer et al. 2017). They are rarely investigated, even though these parameters can be experimentally evaluated.

Resistant clone targeting and selection of drug-susceptible bacteria

Pre-antibiotic condition (at least partially) might be ultimately restored by putting successful efforts to select and enhance the antibiotic-susceptible bacterial reservoir. This can be achieved somehow by creating more favorable conditions for susceptible organisms those resistant to antibiotics. Among several strategies and approaches may include the development of antibiotic hybrids, which can be highly selective in killing a group of organisms resistant to antibiotics. One of the examples includes the bactericidal molecule, which gets activated only upon the particular interaction with the bacteria capable of producing a β-lactamase. Interestingly, screening methods have shown the possibility of uncovering compounds specifically targeting resistant bacteria and often specifically targeting proteins or mechanisms and pathways related to resistance, such as an antibody–drug conjugate that detects an antigen-specific and unique to the surface of the antibiotic-resistant organism (Mariathasan and Tan 2017; Stone et al. 2016). In the long run, innovative antibody–drug carrier systems may confer potency to otherwise inactive antibiotics, such as poorly soluble ones, and then deliver them to the antibody-targeted bacteria in a strain-specific manner (Obeid et al. 2017). One more solution is to evaluate therapies that exclusively reduce the fitness of antibiotic-resistant bacteria, providing a substantive and continuous disadvantage that inevitably leads to their extinction.

Interestingly, some of the "high-risk clones" substantially show good adaptation and include highly transmissible commensals that are resistant to antibiotics in animals and humans. Genetic engineering can be used in antibiotic-resistant clones of bacteria to restore drug susceptibility potential, but it is also a risky ecological approach. Subsequently, CRISPR-Cas or other genome editing systems may be carried out to transform them (resistant clones) into strains that show susceptibility to antibiotics (Goren et al. 2017).

Role of nanotechnology in combating antibiotic resistance

Currently, available antibiotic agents have side effects and face immense resistance from the target microorganism, which has given rise to severe clinical concern (Bhat et al. 2022; Mir et al. 2021; Sheikh et al. 2021). Sub-therapeutic doses of antibiotics at the site of action frequently encourage the establishment of MDR strains of microorganisms, including *M. tuberculosis*. Drug delivery technologies based on nanotechnology provide attractive new opportunities for treating various bacterial infections (tuberculosis), including therapy and diagnosis (Minakshi et al. 2019; Mir et al. 2021). These delivery systems are often made up of nanosized carriers that can be modulated to have various structural or functional properties. High loading capacity, high stability, feasibility for delivery and administration by multiple routes (i.e., intravenous, oral, and inhalation), and relative higher bioavailability are some of the technological advancements of these carriers. Nanocarriers can be altered to have regulated release or stimuli-response release attributes. Further, a nanotechnology-based carrier system can tackle low bioavailability, high systemic clearance, and anatomical barrier issues common with conventional drug delivery (Nasiruddin et al. 2017). Therefore, nanotechnology may help lessen the risks, including the emergence and development of drug-resistant strains of microbes (Fig. 2).

Role of bioinformatics tools for monitoring antimicrobial resistance (AMR)

Monitoring antimicrobial resistance is equally significant as detecting it. In this age of globalization, AMR-related components can quickly drive across borders, increasing the prevalence of AMR on a global scale. For instance, the detection of carbapenemase-resistant *Enterobacterales* (CPE) has dramatically increased over the last few years (Weber et al. 2013), both in hospital and community settings (Azad et al. 2014). Similarly, the use of antibiotics in animals has increased significantly over the past several decades, resulting in the spread of antibiotics into the environment and an increase in the prevalence of AMR in zoonotic agents. The establishment of tools for tracking global AMR has therefore become more significant

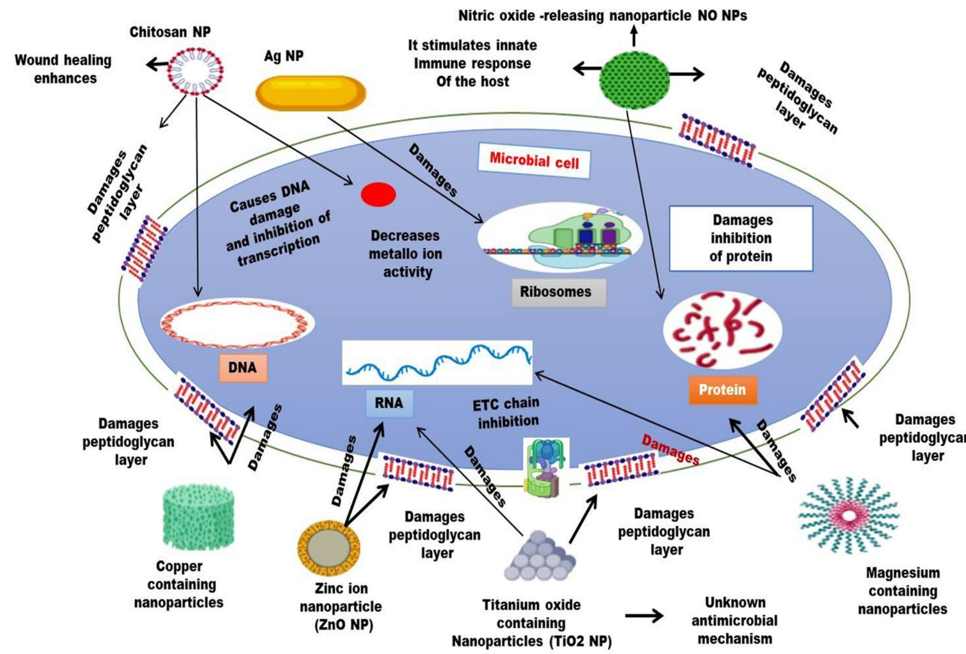

Fig. 2 Nanobiotechnology based drug delivery to combat microbial resistance.Target specific delivery of antimicrobials to effeiciently carry the antimicrobial effect and leading to cell damage in the pathogen

because these tools may offer pertinent information to help healthcare professionals develop stewardship programs and public health initiatives. The European Antimicrobial Resistance Surveillance Network (EARS-Net), Europe's largest AMR surveillance system, collects comparable, representative, and accurate AMR data, analyzing temporal and spatial patterns of AMR, providing evidence of unexpected events, and encouraging immediate mitigation strategies (Bronzwaer et al. 1999).

Furthermore, whole genome sequencing (WGS) enables the detection of AMR because it generates vast amounts of data that will later be analyzed using bioinformatic pipelines. Doing so will make it easier to describe microbial populations and spot patterns in antimicrobial resistance. WGS is being incorporated into clinical and public health settings, but its use has so far mainly been focused on identifying and tracking outbreaks. In a recent survey on the current epidemic of carbapenem-resistant *Klebsiella pneumoniae* (CRKp) in Europe, David et al. showed that acquisition of carbapenemase is the leading cause of carbapenem resistance in *K. pneumoniae* and that almost 70% of the CRKp isolates were concentrated in just four clonal lineages. In addition, analysis of the genomic sequences even showed that the degree of carbapenem resistance is related to the highest transmissibility in hospital settings (David et al. 2019). The observational study conducted by Harris et al. on clinical isolates of *Neisseira gonorrhoeae* from the European Gonococcal Antimicrobial Surveillance Programme is another illustration of using WGS in AMR monitoring (Harris et al. 2018). They revealed that WGS is the best tool for molecular epidemiology because it can easily and rapidly analyze genomic-phylogenetic relationships, identify mixed infections, predict antimicrobial resistance, and provide a realistic picture of the circulating *N. gonorrhoeae* strains (Harris et al. 2018). It was the development of a database containing epidemiological, phenotypic, and genotypic data that can be used for AMR prediction, phylogenetic clustering, and molecular typing (Harris et al. 2018). The development of such databases should be a top priority for the control of the spread of antimicrobial resistance (AMR), as they permit the storage of multiple resistance profiles and genomic data from microorganisms of high epidemiological interest, such as methicillin-resistant *Staphylococcus aureus*, MDR and XDR *Acinetobacter baumannii* or *Pseudomonas aeruginosa*.

Role of inhibitors and CRISPR-Cas gene editing to combat antimicrobial resistance

Enzyme inhibitors

Enzyme inhibitors are synthetic entities with a low subatomic weight that can irreversibly or reversibly limit the catalytic activity of enzymes. This is illustrated by the cholinesterase (ChE) and monoamine oxidase (MAO) inhibitors, which are utilized for a variety of pharmaceutical applications (Hassan et al. 2022). Enzymes are valid targets for therapeutic agents because altering the chemical activity of an enzyme has been shown to significantly affecting disease progression. Indeed, 47% of all the current drugs inhibit target enzymes, even with the improvement in the usage of medicines for the receptors to adjust the signals from outside the cell (Hassan et al. 2022). Several antibiotics carry the inhibition of enzymes, and numerous bacterial enzymes are substantially involved in developing antibiotic resistance. Resistance to these antibiotic drugs occurs when the target enzymes undergo structural modifications or modifications in the target elements affected by antibiotics. Antibiotics usually target enzymes involved in cell wall biosynthesis, replication of nucleic acids, and other metabolites. For example, penicillin-binding proteins (PBPs) are constituents of bacterial cell walls and play a key role in peptidoglycan synthesis. PBPs also catalyse the transpeptidase reaction and transglycosylation, which result in peptide chain crosslinking and elongation. PBPs are a significant target for the majority of antibiotics currently in use. These antibiotics inhibit the biosynthesis of the bacterial cell wall by acting as competitive inhibitors of PBPs (Saraiva et al. 2022). Type II topoisomerases (DNA gyrase and topoisomerase IV), enzymes regulating DNA supercoiling during replication and transcription, are a few more bacterial targets. Antibiotics derived from quinolones, which bind covalently to the active sites of these enzymes and impede replication and transcription, target these enzymes also (Das et al. 2022; Saraiva et al. 2022). Clavulanic acid, a semi-synthetic compound widely used in combination with β-lactam antibiotics to inactivate β-lactamases involved in the breakdown of β-lactam antibiotics, is a classic example of a β-lactamase inhibitor. The inhibitor consists of a β-lactam ring and represents a suicide inhibitor of β-lactamases. Recently, numerous other compounds with inhibitory activity against antimicrobial-resistant enzymes have been explored as potential natural or recombinant weapons in the fight against clinically relevant antibiotic-resistant pathogens (Perbandt et al. 2022; Tooke et al. 2019).

Quorum-sensing inhibitors

Quorum-sensing and biofilm formation are two essential characteristics of bacterial species that increase their chances of survival in adverse environments. One of the most essential means of intracellular communication in bacteria is the quorum-sensing approach. Pathogen quorum-sensing (QS) signaling systems are essential regulators of virulence factor expression and represent highly appealing targets for developing novel therapeutics. A quorum-sensing inhibitor (QSI) is a small molecule with the ability to reduce quorum-sensing regulated gene expression effectively. The quorum-sensing inhibitor must be stable, and the molecules it uses to withstand multiple metabolic attacks. Quorum-sensing inhibitors must also be targeted toward a specific QS system to be effective. Secondary signals are generated when the QS signaling agent interacts with its specific receptor and reaches threshold levels. These secondary signals then bind to multiple promoter regions, resulting in changes in target gene expression in response to the signaling molecules. This implies that QS signaling can be inhibited in three distinct ways. (i) Preventing the synthesis of signal molecules, (ii) the breakdown of signaling molecules by enzymes and (iii) preventing the signaling molecule from interacting with the receptor molecule, which prevents the generation of secondary signals that affect gene expression. Disruption of the regulation circuit can occur in either scenario. Various natural and synthetic molecules can interfere with quorum sensing, and these molecules are currently being investigated in experimental models with promising results. Several different classes of compounds with potential quorum-sensing inhibitory effects have been identified (Saeki et al. 2020). Whether quorum sensing inhibitors will ever help prevent infections is still being debated. It is unclear whether QSIs offers new hope in the ongoing battle against multi-antibiotic-resistant bacteria, but the results appear promising. QSI compounds inhibit the expression of many virulence factors but also make biofilm bacteria more susceptible to conventional antimicrobial treatments. This suggests that a combination treatment of QSI and antibiotics may be beneficial.

CRISPR-Cas gene editing

Clustered regularly interspaced short palindromic repeats (CRISPR)-cas is a unique adaptive immune feature found in bacteria and archaea that protects them from bacteriophage invasion (Barrangou et al. 2007). Bacterial genomes are reprogrammed to include short sequences from bacteriophage or plasmid DNA termed spacers; the guide RNAs from the spacers will be used by the Cas protein machinery to target the invading nucleic acid with the same sequence (Pursey et al. 2018; Shabbir et al. 2019). CRISPR-Cas systems are divided into two classes: class I includes types I, III, and IV, and class II includes types II, V, and VI (Pursey et al. 2018). Numerous Cas proteins in Class I carry out the recognition and cleavage of DNA. In contrast, a single multi-domain Cas protein is responsible for the recognition and cleavage of the DNA in Class II (Pursey et al. 2018). The latter (type II CRISPR-Cas9 system) has and revolutionized molecular biology during the last decade and is employed in prokaryotes and eukaryotic genome editing. Several studies have shown that CRISPR-Cas9 can be used to selectively remove AMR genes from bacterial populations (Bikard et al. 2014; Citorik et al. 2014). It has been demonstrated that conjugative plasmids and phagemids can deliver CRISPR-Cas for selective targeting of AMR genes in plasmids and chromosomes, respectively (Bikard et al. 2014). AMR genes are removed from bacteria, which results in the bacterium being more sensitive to antibiotics (Citorik et al. 2014). It is possible to kill certain bacteria using CRISPR/Cas9 phagemids in vivo (Bikard et al. 2014; Citorik et al. 2014) (Fig. 3). CRISPR-Cas delivery via temperate phages may confer a selective advantage in re-sensitized bacteria (Yosef et al. 2015). It has been demonstrated that a synthetic biology technique based on manipulating phage genomes in *Saccharomyces cerevisiae* may be used to modify the phage host spectrum (Ando et al. 2015). A successful non-viral editing technique employing nanosized CRISPR complexes (polymer-derivatized Cas9 complexed with single guide RNA) has been devised to circumvent the loading and packing efficiency in phage-based delivery. Nanosized CRISPR complexes can successfully target the mec-A methicillin-resistance gene in methicillin-resistant *S. aureus* (Kang et al. 2017). Regardless of the potential, there are still significant challenges ahead in applying the CRISPR-Cas9 system to AMR genes. These include changes to the microbial population following the eradication of AMR bacteria, resistance to anti-CRISPR genes, a narrow host range of CRISPR-Cas vectors and legislation (Pursey et al. 2018).

Role of stem cell-derived antimicrobial peptides

For several years, mesenchymal stem cells (MSCs) have been exhaustively investigated to develop efficient and promising therapeutic products to combat varied chronic ailments. In this direction, the potential of MSCs to control excessive inflammation to promote immunomodulation and tissue healing has proven to be very promising (Harman et al. 2017). More recent investigation has observed that human MSCs produce several factors which exhibit the properties and act like antimicrobial peptides (AMPs). This property has enabled AMPs to promote bacterial growth inhibition in several ways, including preventing the bacteria from their cell wall biosynthesis (Alcayaga-Miranda et al. 2017). This hints that the secretome released by MSCs, which considerably reduces bacterial infections, including antibiotic-resistant MRSA, represents a potential approach or supportive treatment for various related illnesses in the near future. One such study has observed an antibacterial effect in a conditioned medium comprising AMPs, such as hepcidin, lipocal, and LL-37, obtained from human bone marrow (BM) and umbilical-cord MSCs (hUCMSCs), against drug-resistant pathogens (clinical) such as *Klebsiella pneumonia*, *S. aureus* and *E. coli* (McCarthy et al. 2020). Similarly, hUMSCs also displayed inhibitory properties when used against *P. aeruginosa* (imipenem resistant) isolated from human infants (Ren et al. 2020).

Conclusion

In all parts of the world, antibiotic resistance is high. Antibiotic resistance presents the greatest ever challenge to human and animal health. Hence, some critical steps are required to reduce the risk posed by the naturally occurring antimicrobial-resistant genes. Diverse strategies need to be initiated to circumvent antibiotic resistance, including reducing the use of drugs that enhance the mutation in bacteria, fungi, and other pathogens and using those drugs that grossly control the HGT. Moreover, using other potential substances or molecules like biological peptides can serve as alternative antimicrobials compared to the ones used traditionally. In addition, there is a need to develop high-efficacy drug candidates with novel targets in the host and less capability to develop resistance. In this context, the development of antibiotic hybrids can selectively kill a specific group of organisms resistant to other antibiotics. One can also employ antibiotic combination therapies using weak antibiotics in combination with enhancers like adjuvants to avoid the development of antibiotic resistance further. The disparity in the availability of appropriate data is a significant obstacle in resolving antibiotic resistance, which restricts the overall evaluation

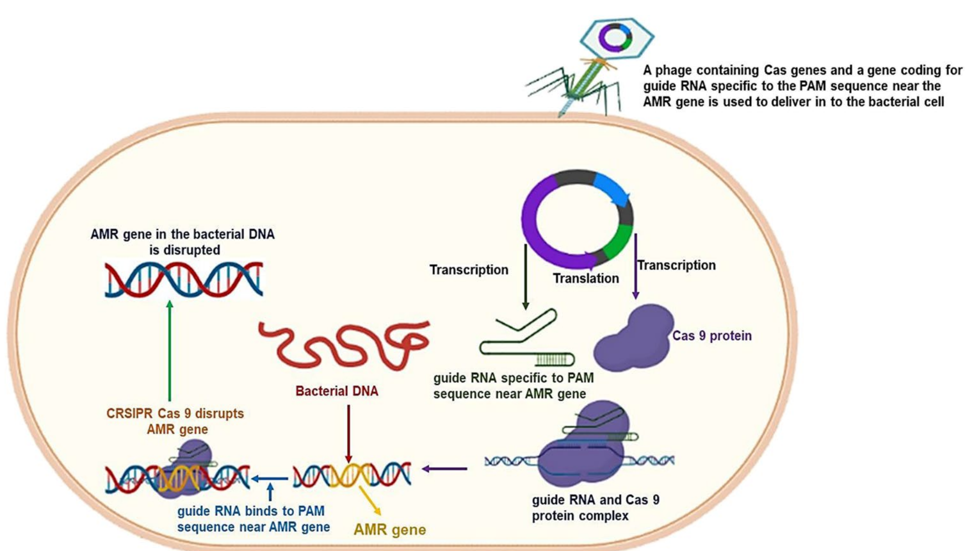

Fig. 3 Detailed mechanism of genome editing approach of CRISPR-Cas system in combating the rising scenario of antimicrobial resistance in micobial pathogens

of the disease burden. The lack of advancement in new technology, instruments, drugs, and studies significantly affects the early diagnosis and management of antibiotic resistance. Various successful interventions, including educational and creative awareness campaigns, have been implemented over time to address the issue of AMR development. Still, successful implementation of such policies and programs needs the coordination and follow-up of patients, their families, clinicians, the government, and various other organizations. Antimicrobial resistance can be combated by appropriate use of antibiotics under the principles of antimicrobial stewardship, the collection of epidemiological data, comprehensive interventions from all walks of life (public, patients, and related health professionals), and the avoidance of antibiotic use in food animals and personal hygiene.

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Declarations

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